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MATH-211

1. Find the domain of the real valued function given by
- $$f(x) = \frac{1}{\sqrt{(1-x)(2-x)}}$$

When we put $x=1, 2$ than $f(x)$ will be undefined

So, domain of function is set of real number except $x \in [1, 2]$

✓ $x \in (-\infty, 1) \cup (2, +\infty)$ Ans

2. Solve the equation $|x| + |x-1| = 0$ for $x < 0$

As (x) is less than zero so the function is negative.

~~$$(x) + (x-1) = 0$$~~

$$-x - (x-1) = 0$$

$$-x - x + 1 = 0$$

$$-2x + 1 = 0$$

$$-2x = -1$$

✓ $x = \frac{1}{2} = \frac{1}{2}$ Ans

Find derivative of the function $5^{\sin 3x}$ with respect to (x)

$$\left(\because \frac{d}{dx} (a^x) = a^x \ln a \right)$$

$$= 5^{\sin 3x} \ln 5 \frac{d}{dx} (\sin 3x)$$

$$= 5^{\sin 3x} \ln 5 [3 \cos x]$$

$$\checkmark = 3 \ln 5 \cdot 5^{\sin 3x} \cos x \quad \text{Ans.}$$

4. Evaluate the integral $\int x^2 e^x dx$
(Integrating by parts)

Let (I) be the required Integral

$$I = \int x^2 e^x dx$$

$$f(x) = \frac{d}{dx} x^2 = 2x$$

$$g(x) = \int e^x = e^x$$

$$I = \int f(x) g(x) dx = f(x) \int g(x) dx - \int \int g(x) f'(x) dx$$

$$I = x^2 (e^x) - \int 2x e^x \quad \text{--- (1)}$$

Again Integrating by parts

$$I = x^2 (e^x) - 2 \int x e^x \quad \text{--- (2)}$$

$$f(x) = \frac{d}{dx} (x) = 1$$

$$g(x) = \int e^x = e^x + c$$

$$\therefore \int x e^x = x e^x - \int (1) e^x$$

$$\int x e^x = x e^x - e^x \quad \text{--- (2)}$$

Putting eq (2) in eq (1)

$$I = x^2 e^x - 2(x e^x - e^x) + c$$

$$\checkmark \quad I = x^2 e^x - 2 x e^x + 2 e^x + c \quad \text{Ans}$$

5. Evaluate the definite integral

$$\int_{-1}^5 |x-2| dx$$

First we have to solve mode:-

$$|x-2| = \begin{cases} -(x-2) & x < 2 \\ x-2 & x \geq 2 \end{cases}$$

$$= \int_{-1}^2 -(x-2) dx + \int_2^5 (x-2) dx$$

$$= -\int_{-1}^2 x dx + 2 \int_{-1}^2 (1) dx + \int_2^5 x dx - 2 \int_2^5 (1) dx$$

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$$= - \left| \frac{x^2}{2} \right|_{-1}^2 + 2 \left| x \right|_{-1}^2 + \left| \frac{x^2}{2} \right|_2^5 - 2 \left| x \right|_2^5$$

$$= -\frac{1}{2} [4-1] + 2 [2+1] + \frac{1}{2} [25-4] - 2 [5-2]$$

$$= -\frac{3}{2} + \frac{6}{1} + \frac{21}{2} - \frac{6}{1}$$

$$= \frac{-3 + 12 + 21 - 12}{2} = +\frac{18}{2}$$

✓

= 9 Ans.

6. Let the radius of circle be increasing at the rate of 2m/s . How fast is the area of that circle increasing when the radius of the circle is 40m ?

Let r and A be the radius and Area of circle respectively.

than $\therefore A = \pi r^2$

$$\frac{dA}{dt} = \frac{d}{dt} (\pi r^2)$$

$$\frac{dA}{dt} = \pi (2r) \frac{dr}{dt}$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} \quad \text{--- (1)}$$

Given that $\therefore r = 40\text{m}$

$$\frac{dr}{dt} = 2\text{ms}$$

Putting values in eq (1)

$$\frac{dA}{dt} = 2\pi \times 40 \times 2$$

$$\frac{dA}{dt} = 160\pi \text{m}^2/\text{s} \text{ Ans.}$$

✓

Question No # 2:- (Long)

1. Examine whether the given function is continuous at $x=0$

$$f(x) = \begin{cases} (1+3x)^{\frac{1}{3x}} & , x \neq 0 \\ e^2 & , x = 0 \end{cases}$$

(C-1)

$$f(0) = e^2 \quad (f \text{ is defined})$$

(C-2) $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (1+3x)^{\frac{1}{3x}}$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (1+3x)^{\frac{1}{3x}}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left[(1+3x)^{\frac{1}{3x}} \right]^3$$

$$\lim_{x \rightarrow 0} f(x) = e^3 \quad \left(\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \right)$$

(C-3)

$$\lim_{x \rightarrow 0} f(x) \neq f(0)$$

so, f is not continuous

2. Find $\frac{dy}{dx}$ by implicit differentiation for the curve

$$\tan^{-1}\left(\frac{y}{x}\right) + yx^2 = 1$$

Differentiate w.r.t "x"

(Applying product rule)

$$\frac{d}{dx} \left(\tan^{-1}\left(\frac{y}{x}\right) + y(2x) + x^2 \frac{dy}{dx} \right) = \frac{d}{dx} 1$$

(Applying quotient rule)

$$\frac{1}{1 + \frac{y^2}{x^2}} \left[\frac{x \frac{dy}{dx} - y(1)}{x^2} \right] + 2xy + x^2 \frac{dy}{dx} = 0$$

$$\frac{1}{\frac{x^2 + y^2}{x^2}} \left[\frac{x \frac{dy}{dx} - y}{x^2} \right] + 2xy + x^2 \frac{dy}{dx} = 0$$

$$\frac{x^2}{x^2 + y^2} \left[\frac{x \frac{dy}{dx} - y}{x^2} \right] + 2xy + x^2 \frac{dy}{dx} = 0$$

$$\frac{x \frac{dy}{dx} - y}{x^2 + y^2} + 2xy + x^2 \frac{dy}{dx} = 0$$

$$\frac{x}{x^2 + y^2} \frac{dy}{dx} - \frac{y}{x^2 + y^2} + 2xy + x^2 \frac{dy}{dx} = 0$$

$$\frac{x}{x^2+y^2} \frac{dy}{dx} + x^2 \frac{dy}{dx} = \frac{y}{x^2+y^2} - 2xy$$

$$\frac{dy}{dx} \left[\frac{x}{x^2+y^2} + x^2 \right] = \frac{y-2xy(x^2+y^2)}{x^2+y^2}$$

$$\frac{dy}{dx} \left[\frac{x + x^2(x^2+y^2)}{\cancel{x^2+y^2}} \right] = \left[\frac{y-2x^3y-2xy^3}{\cancel{x^2+y^2}} \right]$$

$$\frac{dy}{dx} [x + x^4 + x^2y^2] = y - 2x^3y - 2xy^3$$

✓ $\frac{dy}{dx} = \frac{y(1-2x^3-2xy^2)}{x(1+x^3+xy^2)}$ Ans

Evaluate $\int \frac{1}{3\sin x + 4\cos x} dx$

Suppose -

$$3 = r \cos \theta \quad \dots \dots (i)$$

$$4 = r \sin \theta \quad \dots \dots (ii)$$

Adding equation (i) & (ii):-

$$r \cos \theta + r \sin \theta = 3 + 4$$

(Squaring both sides)

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta = (3)^2 + (4)^2$$

$$r^2 (\cos^2 \theta + \sin^2 \theta) = 9 + 16$$

$$\text{As } (\sin^2 \theta + \cos^2 \theta = 1)$$

$$\sqrt{r^2} = \sqrt{25}$$

$$r = \pm 5$$

Dividing eq (ii) & (iii)

~~$$\frac{\sin \theta}{\cos \theta} = \frac{4}{3}$$~~

~~$$\frac{\sin \theta}{\cos \theta} = \frac{4}{3}$$~~

$$\frac{\sin \theta}{\cos \theta} = \frac{4}{3} \quad (\text{As } \frac{\sin \theta}{\cos \theta} = \tan \theta)$$

$$\tan \theta = \frac{4}{3}$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right) \dots \dots (3)$$

$$I = \int \frac{dx}{r \sin \theta \cos x + r \cos \theta \sin x}$$

$$I = \int \frac{dx}{r (\sin \theta \cos x + \cos \theta \sin x)}$$

$$I = \frac{1}{r} \int \frac{dx}{\sin \theta \cos x + \cos \theta \sin x}$$

$$\text{As } (\sin \alpha + \beta = \sin \alpha \cos \beta + \cos \alpha \sin \beta)$$

$$I = \frac{1}{r} \int \frac{dx}{\sin(x + \theta)}$$

∴ Substituting value of r

$$I = \frac{1}{5} \int \frac{dx}{\sin(x+\theta)}$$

∴ As $\frac{1}{\sin x} = \operatorname{cosec} x$

$$I = \frac{1}{5} \int \operatorname{cosec}(x+\theta) dx$$

✓ Since $\int \operatorname{cosec} x dx = \log \tan\left(\frac{x}{2}\right) + c$

or

$$\int \operatorname{cosec} x dx = \log |\operatorname{cosec} x - \cot x| + c$$

✓ $I = \frac{1}{5} \log \tan \frac{x+\theta}{2} + c$ Ans.

Now substitute value of (θ)
from eq (3) :-

$$I = \frac{1}{5} \log \tan \left[\frac{x}{2} + \frac{\tan^{-1} \frac{4}{3}}{2} \right] + c$$

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1. Find the domain of real valued function given by $f(x) = \sqrt{4-x^2}$

$$4 - x^2 \geq 0$$

$$+ x^2 \geq 4$$

$$x^2 \geq 4$$

$$\sqrt{x^2} \geq \sqrt{4}$$

$$x = \pm 2$$



✓

$$x \in [-2, 2] \quad \text{Ans.}$$

2. Solve the equation $|2x| + |x-3| = 0$
for $x < 0$

As (x) is less than zero

So, function is negative

$$-(2x) - (x-3) = 0$$

$$-2x - x + 3 = 0$$

$$-3x + 3 = 0$$

$$-3x = -3$$

$$x = \frac{-3}{-3}$$

✓

$$x = 1 \quad \text{Ans}$$

3. Find derivative of the function $e^{x \cos 5x}$ with respect to "x"

$$e^{x \cos 5x}$$

Differentiate w.r.t "x"

$$= e^{x \cos 5x} \frac{d}{dx} x \cos 5x$$

$$= e^{x \cos 5x} \left[x \frac{d}{dx} \cos 5x + \cos 5x \frac{d}{dx} x \right]$$

$$= e^{x \cos 5x} \left[x (-\sin 5x) \frac{d}{dx} (5x) + \cos 5x \right]$$

$$= e^{x \cos 5x} \left[x (-\sin 5x) (5) + \cos 5x \right]$$

$$\checkmark = e^{x \cos 5x} [-5x \sin 5x + \cos 5x] \text{ Ans.}$$

4. Evaluate the integral $\int x^2 \ln x dx$

$$\int x^2 \ln x dx$$

$$f(x) = \frac{d}{dx} \ln x = \frac{1}{x}$$

$$g(x) = \int x^2 = \frac{x^3}{3} + C$$

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$$f(u)g(u) = f(u) \int g(u) - \iint g(u) \cdot f'(u) dx$$

$$I = \ln x \left(\frac{x^3}{3} \right) - \int \frac{x^3}{3} \cdot \frac{1}{x} dx$$

$$I = \ln x \left(\frac{x^3}{3} \right) - \int \frac{x^3}{3x} dx$$

$$I = \frac{\ln x \times x^3}{3} - \frac{1}{3} \int \frac{x^3}{x} dx$$

$$I = \frac{\ln(x) \times x^3}{3} - \frac{1}{3} \left[\frac{x^2}{2} dx \right]$$

$$I = \frac{\ln x \times x^3}{3} - \frac{1}{3} \left[\frac{x^3}{3} + c \right]$$

$$I = \frac{\ln x \times x^3}{3} - \left[\frac{1}{9} \times x^3 \right] + c$$

$$I = \frac{\ln x \times x^3}{3} - \frac{x^3}{9} + c \quad \text{Ans.}$$

5. Evaluate the definite Integral $\int_0^{\pi} \sin 3x \cos 5x dx$

Since $(2 \sin A \cos B = \sin(A+B) + \sin(A-B))$

Multiplying and Dividing by (2)

$$= \int_0^{\pi} \frac{1}{2} (2 \sin 3x \cos 5x) dx$$

$$= \int_0^{\pi} \frac{1}{2} [\sin(3x+5x) + \sin(3x-5x)] dx$$

$$= \int_0^{\pi} \frac{1}{2} [\sin(8x) + \sin(-2x)] dx$$

$$= \int_0^{\pi} \frac{1}{2} [\sin 8x - \sin 2x] dx$$

$$= \frac{1}{2} \int_0^{\pi} \sin 8x - \frac{1}{2} \int_0^{\pi} \sin 2x$$

$$= \frac{1}{2} \left| -\frac{\cos 8x}{8} \right|_0^{\pi} - \frac{1}{2} \left| -\frac{\cos 2x}{2} \right|_0^{\pi}$$

$$= - \left| \frac{\cos 8x}{16} \right|_0^{\pi} + \left| \frac{\cos 2x}{4} \right|_0^{\pi}$$

$$= - \left(\frac{\cos 8\pi - \cos 0^\circ}{16} \right) + \left(\frac{\cos 2\pi - \cos 0^\circ}{4} \right)$$

$$- \left(\frac{1-1}{16} \right) + \left(\frac{1-1}{4} \right)$$

$$- \left(\frac{0}{16} \right) + \left(\frac{0}{4} \right) = 0 + 0 = 0$$

6. Let the radius of the circle be increasing at the rate 2m/s . How fast is the area of that circle increasing when the radius of circle is 40m ?

Let r and A be the radius And Area of the circle, respectively

$$\text{Then :- } A = \pi r^2$$

$$\frac{dA}{dt} = \frac{d}{dt} (\pi r^2)$$

$$\frac{dA}{dt} = 2\pi r \left(\frac{dr}{dt} \right)$$

It is given that :-

$$r = 2\text{m/s}$$

$$\frac{dr}{dt} = 40\text{m}$$

$$\frac{dA}{dt} = 2\pi \times 2 \times 40$$

$$\frac{dA}{dt} = 160\pi \text{ m}^2/\text{s}$$

Question No #2 :- (LONG) :-

1. Examine the given function is continuous at $x=0$

$$f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$$

Condition - 1 :-

$$f(0) = 2 \quad (f \text{ is defined})$$

Condition - 2 :-

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$\lim_{x \rightarrow 0} f(x) = 1 \quad \left(\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

condition - 3 :-

$$f(0) \neq f(x)$$

(So, $f(x)$ is not continuous at $x=0$) **Ans.**

2. Find $\frac{dy}{dx}$ by implicit differentiation for the curve $y \sin' x - x \tan' y = 1$

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$$y \sin^{-1}(x) - x \tan^{-1}(y) = 1$$

($\therefore$ using product rule)

$$y \frac{1}{\sqrt{1-x^2}} + \sin^{-1}x \frac{dy}{dx} - \frac{x}{1+y^2} \frac{dy}{dx} - \tan^{-1}y = 0$$

$$\frac{y}{\sqrt{1-x^2}} + \sin^{-1}x \frac{dy}{dx} - \frac{x}{1+y^2} \frac{dy}{dx} - \tan^{-1}y = 0$$

$$\sin^{-1}x \frac{dy}{dx} - \frac{x}{1+y^2} \frac{dy}{dx} = \tan^{-1}y - \frac{y}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} \left[\sin^{-1}x - \frac{x}{1+y^2} \right] = \tan^{-1}y - \frac{y}{\sqrt{1-x^2}}$$

$$\checkmark \quad \frac{dy}{dx} = \frac{\tan^{-1}y - \frac{y}{\sqrt{1-x^2}}}{\sin^{-1}x - \frac{x}{1+y^2}} \quad \text{Ans.}$$

3. Show that the function $f(x) = |x|$ is continuous but not differentiable at $x=0$?

We have $f(x) = |x|$ if $x=0$

i) $f(0) = 0$

ii) if $x > 0$ then $\lim_{h \rightarrow 0} f(a+h)$

$$= \lim_{h \rightarrow 0} (0+h) = 0$$

$$\text{iii) if } x < 0 \text{ then } \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} f(0-h) = 0$$

$$\lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} f(a-h) = f(0)$$

So, the $f(x)$ is continuous at $x=0$

For differentiation:-

$$\text{i) } f(0) = 0$$

$$\begin{aligned} \text{ii) if } x > 0 \text{ then } \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ = \lim_{h \rightarrow 0} \frac{0+h-0}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1 \end{aligned}$$

$$\text{iii) if } x > 0 \text{ then } \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{+h}{-h} = -1$$

$$\lim_{h \rightarrow 0} R.H.L \neq L.H.L \neq f'(x)$$

So, $f(x)$ is not differentiable at $x=0$

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2. Solve $|x| + |x-1| > 1$

$$\pm(x) \pm (x-1) > 1 \quad (\text{Four cases are possible})$$

(1) $-(x) + (x-1) > 1$

$$\bullet -x + x - 1 > 1$$

$$-1 > 1 \quad (\text{This is not possible})$$

(2) $(x) - (x-1) > 1$

$$x - x + 1 > 1$$

$$1 > 1 \quad (\text{This is not possible})$$

(3) $-(x) - (x-1) > 1$

$$-x - x + 1 > 1$$

$$-2x > 1 - 1$$

$$-2x > 0$$

$$x < \frac{0}{-2}$$

$$x < 0$$

Solution this is $(-\infty, 0)$

(4) $x + x - 1 > 1$

$$2x > 1 + 1$$

$$2x > 2$$

$$x > \frac{2}{2} = 1$$

solution of this is $(1, \infty)$

✓ S.S = $\{(-\infty, 0) \cup (1, \infty)\}$ Ans

ii) Evaluate

$$\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$$

$$= \lim_{x \rightarrow 0} \frac{\sin ax \times \frac{ax}{ax}}{\sin bx \times \frac{bx}{bx}}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin ax}{ax} \times ax}{\frac{\sin bx}{bx} \times bx}$$

$$= \frac{a}{b} \frac{\lim_{x \rightarrow 0} \frac{\sin ax}{ax}}{\lim_{x \rightarrow 0} \frac{\sin bx}{bx}}$$

$$\checkmark = \frac{a}{b} \frac{(1)}{(1)} = \frac{a}{b} \quad \text{Ans.}$$

iii) Find $\frac{dy}{dx}$ if $x^3 + y^3 - 3axy = 0$

Differentiate w.r.t "x"

$$\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) - 3a \frac{d}{dx}(xy) = 0$$

Applying differentiation separately.

$$= 3x^2 + 3y^2 \cdot \frac{dy}{dx} - 3a \left[x \frac{dy}{dx} + y \right] = 0$$

$$3x^2 + 3y^2 \cdot \frac{dy}{dx} - 3a \left[x \frac{dy}{dx} + y \right] = 0$$

$$3 \left[x^2 + y^2 \cdot \frac{dy}{dx} - ax \cdot \frac{dy}{dx} - ay \right] = 0$$

$$x^2 + y^2 \cdot \frac{dy}{dx} - ax \cdot \frac{dy}{dx} - ay = 0$$

$$\frac{dy}{dx} [y^2 - ax] = ay - x^2$$

$$\checkmark \quad \frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax} \quad \text{Ans.}$$

4. Evaluate $\int x \sec^2 x dx$

Let (I) be the required
Integral

$$I = \int x \sec^2 x dx$$

$$f(x) = \frac{d}{dx}(x) = 1$$

$$g(x) = \int \sec^2 x = \tan x + c$$

$$I = x \tan x - \int (\tan x)$$

$$I = x \tan x - \int \frac{\sin x}{\cos x} dx$$

$$I = x \tan x + \int \frac{-\sin x}{\cos x} dx$$

$$\text{Let } t = \cos x$$

$$dt = -\sin x$$

$$I = x \tan x + \int \frac{1}{t} dt$$

$$I = x \tan x + \ln |t| + c$$

$$I = x \tan x + \ln |\cos x| + c$$

5. Evaluate $\int_1^2 \frac{x^2 + 1}{x+1} dx$

Adding and subtracting (1) in Numerator

$$\int_1^2 \frac{x^2 + 1 - 1 + 1}{x+1} dx$$

$$\int_1^2 \left[\frac{x^2 - 1}{x+1} + \frac{2}{x+1} \right] dx$$

$$\int_1^2 \left[\frac{(x+1)(x-1)}{x+1} + \int_1^2 \frac{2}{x+1} \right] dx$$

$$\int_1^2 (x-1) dx + 2 \int_1^2 \left(\frac{1}{x+1} \right) dx$$

Integrating separately.

$$\int_1^2 x dx - \int_1^2 1 dx + 2 \int_1^2 \frac{1}{x+1} dx$$

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$$\left| \frac{x^2}{2} \right|_1^2 - |x|_1^2 + 2 \left| \ln|x+1| \right|_1^2$$

Applying upper & lower limits

$$\frac{(4)^2}{2} = 2 + 2 \ln|2+1| - \left(\frac{(1)^2}{2} - 2 + 2 \ln|1+1| \right)$$

$$\cancel{2} - \cancel{2} + 2 \ln|3| - \left(\frac{1}{2} - \frac{2}{1} + 2 \ln|2| \right)$$

$$2 \ln|3| - \left(\frac{+1-2}{2} + 2 \ln|2| \right)$$

$$2 \ln|3| - \left(\left(-\frac{1}{2} \right) + 2 \ln|2| \right)$$

$$2 \ln|3| + \frac{1}{2} + 2 \ln|2|$$

$$2 \ln|3| + 2 \ln|2| + \frac{1}{2}$$

$$\checkmark \quad 2 \ln \left[\frac{3}{2} \right] + \frac{1}{2} \quad \underline{\text{Ans}}$$

Question No #2 :- (LONG)

i) solve $\frac{x^2 - 2}{1 - 2x} > 1$

Let the associated equation be:-

$$\frac{x^2 - 2}{1 - 2x} = 1$$

$$x^2 - 2 = 1 - 2x$$

$$x^2 + 2x = 1 + 2$$

$$x^2 + 2x = 3$$

$$x^2 + 2x - 3 = 0$$

$$x^2 + 3x - x - 3 = 0$$

$$x(x+3) - 1(x+3) = 0$$

$$(x-1)(x+3) = 0$$

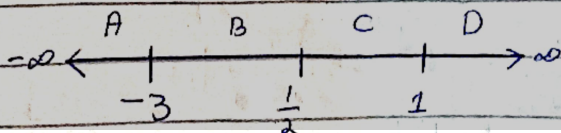
$x = 1, -3$ are boundary numbers

$$1 - 2x = 0$$

$$-2x = -1$$

$$x = \frac{1}{2}$$

$x = \frac{1}{2}$ is free boundary number



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For region A:-Put $x = -5$

$$\frac{(-5)^2 - 2}{1 - 2(-5)} > 1$$

$$\frac{25 - 2}{1 + 10} > 1$$

$$\frac{23}{11} > 1 \quad (\text{True})$$

For region B:-Put $x = 0$

$$\frac{(0)^2 - 2}{1 - 0} > 1$$

$$\frac{-2}{1} > 1$$

$$-2 > 1 \quad (\text{False})$$

For region C:-Put $x = 0.9$

$$\frac{(0.9)^2 - 2}{1 - 2(0.9)} > 1$$

$$\frac{0.81 - 2}{1 - 0.81} > 1$$

$$-\frac{1.19}{0.19} > 1 \quad (\text{False})$$

ii)

For region D:-

put $x = 10$

$$\frac{(10)^2 - 2}{1 - 2(10)} > 1$$

$$\frac{100 - 2}{1 - 20} > 1$$

$$\frac{98}{-19} > 1$$

$$-\frac{98}{19} > 1 \quad (\text{False})$$

$$S.S = (-\infty, -3) \cup (\frac{1}{2}, 1)$$

ii) Find $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$

Let $y = a^x - 1$

than $y + 1 = a^x$

$$\ln(1 + y) = \ln a^x$$

$$\ln(1 + y) = x \ln a$$

$$x = \frac{\ln(1 + y)}{\ln a}$$

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} (a^x - 1)$$

$$\lim_{x \rightarrow 0} y = a^0 - 1$$

$$\lim_{x \rightarrow 0} y = 1 - 1 = 0$$

This shows that as $y \rightarrow 0$ so $x \rightarrow 0$ also. So, Limit can be written as:-

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{y \rightarrow 0} \frac{y}{\frac{\ln(a)}{\ln(1+y)}}$$

$$\lim_{y \rightarrow 0} \frac{(\ln a)(y)}{\ln(1+y)} = \lim_{y \rightarrow 0} \frac{\ln a}{\frac{1}{y} \ln(1+y)}$$

$$\lim_{y \rightarrow 0} \frac{\ln a}{\ln(1+y)^{\frac{1}{y}}} = \frac{\ln a}{\ln \left[\lim_{y \rightarrow 0} (1+y)^{\frac{1}{y}} \right]}$$

As we known:-

$$\lim_{y \rightarrow 0} (1+y)^{\frac{1}{y}} = e, \text{ In } e = 1$$

$$\checkmark = \frac{\ln a}{\ln e} = \frac{\ln a}{1} = \ln a \text{ Ans}$$

3. Differentiate w.r.t "x"

i) $y = e^{ax} \cos(b \arctan x)$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} \left(e^{ax} \times \cos(b \arctan x) \right) \\
 &= e^{ax} \frac{d}{dx} \cos(b \arctan x) + \cos(b \arctan x) \frac{d}{dx} e^{ax} \\
 &= e^{ax} \left(-\sin \frac{d}{dx} b \tan^{-1} x \right) + \cos(b \tan^{-1} x) e^{ax} \\
 &= e^{ax} \left[-\sin(b \tan^{-1} x) \left(-b \times \frac{1}{1+x^2} \right) \right] + \cos(b \tan^{-1} x) e^{ax} \\
 &= \frac{+ b e^{ax} \sin(b \tan^{-1} x)}{1+x^2} + a e^{ax} \cos(b \tan^{-1} x)
 \end{aligned}$$

ii) $y = x a^x \sinh x$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx} (a^x \times \sinh x) \\
 &= \sinh x \frac{d}{dx} (a^x) + a^x \frac{d}{dx} \sinh x \\
 &= \sinh x \left[a^x \frac{d}{dx} x + x \frac{d}{dx} a^x \right] + a^x \cosh x \\
 &= \sinh x \left[a^x + x a^x \ln a \right] + a^x \cosh x \\
 &= a^x \sinh x + a^x x \ln a \sinh x + a^x \cosh x
 \end{aligned}$$

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Q.4 Evaluate

$$\int -\frac{1}{\sqrt{a^2-x^2}}$$

$$x = a \sin \theta$$

differentiate w.r.t "x"

$$dx = a(\cos \theta d\theta)$$

$$dx = a \cos \theta d\theta \quad \text{---(i)}$$

$$\theta = \sin^{-1}\left(\frac{x}{a}\right) \quad \text{---(ii)}$$

$$\int -\frac{1}{\sqrt{a^2-(a \sin \theta)}} \cdot a \cos \theta d\theta$$

$$\int -\frac{1}{\sqrt{a^2-a^2 \sin^2 \theta}} \cdot a \cos \theta d\theta$$

$$\int -\frac{1}{\sqrt{a^2(1-\sin^2 \theta)}} \cdot a \cos \theta d\theta$$

$$\int -\frac{1}{\sqrt{a^2 \cdot \cos^2 \theta}} \cdot a \cos \theta d\theta$$

$$\int -\frac{1}{a \cdot \cos \theta} \cdot a \cos \theta d\theta$$

$$\int -1 d\theta$$

$$= -\int 1 d\theta$$

Integrating

Putting value of θ from eq (ii):-

$$= -\sin^{-1}\left(\frac{x}{a}\right)$$

✓

$$= -\arcsin\left(\frac{x}{a}\right) \text{ Ans.}$$

$$\int_0^{\pi/6} x \cos x dx$$

Let (I) be required Integral

Integrating by Parts

$$\text{Let } I = \int_0^{\pi/6} x \cos x dx$$

$$f(x) = \frac{d}{dx}(x) = 1$$

$$g(x) = \int_0^{\pi/6} \cos x dx = \sin x + c$$

$$I = x(\sin x) - \int_0^{\pi/6} \sin x dx$$

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$$= x \sin x - (-\cos x)$$

$$= x \sin x + \cos x$$

$$\left| x \sin x \right|_0^{\pi/6} + \left| \cos x \right|_0^{\pi/6}$$

$$= \left(\frac{\pi}{6} \sin \frac{\pi}{6} - 0 \sin 0 \right) + \left(\cos \frac{\pi}{6} - \cos 0 \right)$$

$$= \left(\frac{\pi}{6} \times \frac{1}{2} - 0 \right) + \left(\frac{\sqrt{3}}{2} - 1 \right)$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 \quad \text{Ans.}$$